

in an extremely risk averse way, which manifests in their selling equities. An investment bank, for example, cannot roll over its debt or there are outflows from an asset manager. Both of these situations can similarly raise the aggregate risk aversion of the economy during bad times.

For an asset owner contemplating a large position in equities, heterogeneity implies that she should think not only about how she is different from the market (or the marginal or representative agent) but also what kind of investor she is relative to the full distribution of investors. The behavior of these other investors and how they interact affects asset prices. The main intuition of bad times survives in these heterogeneous agent models—the equity premium is high because some agents (they do not have to be numerous, but when they matter, they matter a lot) find that equity is unattractive for their set of bad times. These may not be bad times for her, and if so, she should hold an above-average allotment of equities.

The problem in applying the insights of heterogeneous agent models to investment policy is that the world is so heterogeneous. It is unclear what dimensions—risk aversion, wealth, loss capacity, leverage capacity, (lack of) liabilities, income, or aversion to downside events—we should emphasize. It is often difficult, if not impossible, to measure cross-sectional characteristics of investor types, which the heterogeneous agent economies predict should be useful candidate factors. Since a representative agent often arises in these models anyway, my advice is to concentrate on the first three explanations of the equity risk premium in terms of considering how you differ from the market. The implication from heterogeneous agent models is that you may interpret the “market” as the “marginal investor” as well as the “average investor.”

4. Equities and Inflation

4.1. STOCKS ARE A BAD INFLATION HEDGE

Equity is a real security, in the sense that it represents a claim on real, productive assets of firms. But it turns out that equities actually are not a real security in the sense that “real” means “inflation adjusted” or the opposite of “nominal.” In fact quite the opposite: equities are bad at hedging inflation risk.¹⁶

That may surprise you but take a look at Figure 8.4, which graphs correlations between inflation and stock returns, excess stock returns relative to T-bills (cash), and T-bill returns from 1926 to 2010. Inflation hedging is about the co-movement of assets with inflation, not about long-run average returns. The correlations are computed over various horizons measured in years on the x -axis. I use log returns and log inflation changes so that the long-horizon returns are sums of the short frequency, one-month returns (see the appendix).

¹⁶ An early reference on the negative relationship between equity returns and inflation is Lintner (1975).